

Day 1 – Machine Learning Ability Assessment

STUDENT PERFORMANCE PREDICTION & ANALYSIS SYSTEM

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Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook

Dataset: Student Performance Prediction Dataset

A machine learning classification project for predicting student academic outcomes.

1. Introduction

The objective of this project is to develop a machine learning system that predicts whether a student is likely to pass or fail based on academic and related performance factors. The project covers data preparation, exploratory data analysis, classification model development, evaluation and prediction for a new student.

Two classification algorithms, Logistic Regression and K-Nearest Neighbors (KNN), were implemented using Scikit-learn and compared using standard classification metrics.

2. Dataset Overview

The final dataset contains 10,000 records and 9 columns. It includes academic, attendance and student-related information used to predict the target variable Passed.

Features: Study Hours per Week, Attendance Rate, Previous Grades, Assignment Score, Midterm Score, Absences, Participation in Extracurricular Activities and Parent Education Level.

The dataset was based on a publicly available student performance dataset and was modified and extended before model development.

3. Data Preprocessing

The dataset was checked for missing values and duplicate records. The final dataset contained no missing values or duplicate rows. Categorical variables were encoded into numerical form, including binary encoding for extracurricular participation and one-hot encoding for parent education level. The target variable was encoded as Pass = 1 and Fail = 0.

The data was divided into training and testing sets using an 80:20 split. StandardScaler from Scikit-learn was then used for feature scaling.

4. Exploratory Data Analysis

EDA was performed to understand the distribution of numerical features and identify patterns in the student data. Histograms were used to examine study hours, attendance, previous grades and other numerical variables.

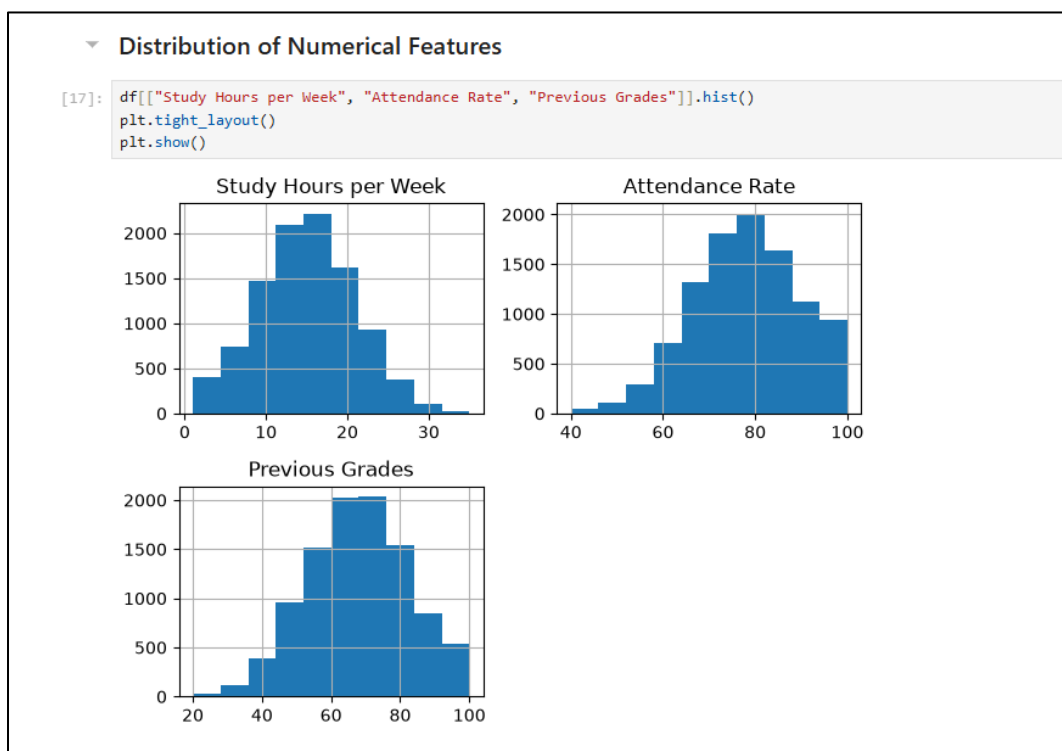


Figure 1: Distribution of Numerical Features

The distributions provide an overview of how the numerical values are spread across the students in the dataset.

5. Machine Learning Model Development

Two classification models were selected and trained using the same training and testing data.

Logistic Regression: A simple and interpretable binary classification algorithm used to predict the Pass or Fail class.

K-Nearest Neighbors (KNN): A distance-based classification algorithm that predicts a class using nearby data points. Feature scaling was applied before training.

6. Model Evaluation

The models were evaluated using accuracy, precision, recall and F1 score. These metrics provide a broader view of classification performance than accuracy alone.

Logistic Regression

Accuracy: 62.90% Precision: 63.28% Recall: 63.91% F1 Score: 63.59%

K-Nearest Neighbors

Accuracy: 60.15% Precision: 60.93% Recall: 59.66% F1 Score: 60.29%

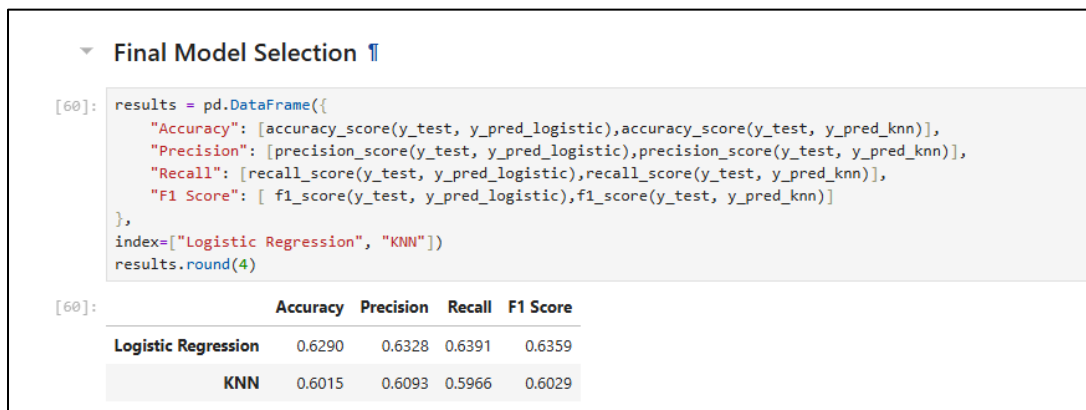


Figure 2: Final Model Selection

Logistic Regression achieved higher accuracy, precision, recall and F1 score than KNN. Based on the overall results, Logistic Regression was selected as the final model.

7. Final Student Prediction

A simple prediction mechanism was implemented using the trained Logistic Regression model. It accepts the same type of information used during model training and returns a Pass or Fail prediction.

Sample Input: Study Hours = 20, Attendance = 85, Previous Grades = 75, Assignment Score = 80, Midterm Score = 78, Absences = 3, Extracurricular Activities = Yes, Parent Education = Bachelor.

Prediction: Pass

```
Final Prediction

[63]: study_hours = float(input("Study hours per week: "))
      attendance = float(input("Attendance rate: "))
      grades = float(input("Previous grades: "))
      assignment_score = float(input("Assignment score: "))
      midterm_score = float(input("Midterm score: "))
      absences = float(input("Absences: "))

      participation = input("Extracurricular activities (Yes/No): ")
      education = input("Parent education (Associate/Bachelor/Doctorate/High School/Master): ")

      student = [[
          study_hours,
          attendance,
          grades,
          assignment_score,
          midterm_score,
          absences,
          1 if participation == "Yes" else 0,
          1 if education == "Associate" else 0,
          1 if education == "Bachelor" else 0,
          1 if education == "Doctorate" else 0,
          1 if education == "High School" else 0,
          1 if education == "Master" else 0
      ]]

      student = scaler.transform(student)

      prediction = logistic_model.predict(student)

      if prediction[0] == 1:
          print("Prediction: Pass")
      else:
          print("Prediction: Fail")

      Study hours per week: 20
      Attendance rate: 85
      Previous grades: 75
      Assignment score: 80
      Midterm score: 78
      Absences: 3
      Extracurricular activities (Yes/No): Yes
      Parent education (Associate/Bachelor/Doctorate/High School/Master): Bachelor
      Prediction: Pass
```

Figure 3: Final Student Prediction

8. Limitations

- The model uses only the features available in the dataset.
- Factors such as motivation, learning environment and teaching quality are not included.
- The prediction does not guarantee the actual academic outcome.
- Model performance may change when applied to a different real-world dataset.

9. Future Improvements

- Add more relevant student performance features.
- Test additional classification algorithms and perform hyperparameter tuning.
- Apply feature selection and cross-validation for more reliable evaluation.
- Develop and deploy a simple web interface for predictions.

10. Project Deliverables

Jupyter Notebook/Python project, dataset or dataset source, EDA results, preprocessing, two classification models, model evaluation, model comparison, sample prediction, README/documentation, screenshots, explanation video and GitHub repository.

11. Conclusion

The Student Performance Prediction & Analysis System demonstrates a complete machine learning classification workflow. The dataset was explored and prepared before model development, and Logistic Regression and KNN were evaluated using multiple classification metrics. Logistic Regression achieved the best overall performance with 62.90% accuracy and was selected as the final model. The project provided practical experience in preprocessing, EDA, classification, model evaluation and prediction of new student data.